Q1: Personal Portfolio Webpage Layout (Flexbox)

index.html

HTML

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <title>Portfolio - Q1</title>
  <link rel="stylesheet" href="style.css">
</head>
<body>
  <header class="header">
    <div class="logo">John Doe</div>
    <nav class="nav-links">
      <a href="#about">About</a>
      <a href="#projects">Projects</a>
      <a href="#contact">Contact</a>
    </nav>
  </header>

  <main class="main-content">
    <section class="about-section" id="about">
      <h2>About Me</h2>
      <p>Hello! I am a web development student learning modern CSS layouts like Flexbox and Grid. I love building responsive websites.</p>
    </section>
```

```
        <section class="image-section">
            <img src="https://via.placeholder.com/300" alt="Portfolio Image">
        </section>
    </main>

    <footer class="footer">
        <p>&copy; 2026 John Doe. All rights reserved.</p>
    </footer>
</body>
</html>
```

style.css

CSS

```
* {
    margin: 0;
    padding: 0;
    box-sizing: border-box;
}

body {
    font-family: Arial, sans-serif;
    display: flex;
    flex-direction: column;
    min-height: 100vh;
}

.header {
```

```
    display: flex;
    justify-content: space-between;
    align-items: center;
    background-color: #333;
    color: white;
    padding: 20px 40px;
}

.nav-links a {
    color: white;
    text-decoration: none;
    margin-left: 20px;
}

.main-content {
    display: flex;
    flex: 1;
    justify-content: center;
    align-items: center;
    padding: 40px;
    gap: 40px;
    background-color: #f4f4f4;
}

.about-section, .image-section {
    flex: 1;
```

```
    padding: 20px;
    background: white;
    border-radius: 8px;
    text-align: center;
}

.footer {
    background-color: #222;
    color: white;
    text-align: center;
    padding: 15px;
    display: flex;
    justify-content: center;
    align-items: center;
}

@media (max-width: 700px) {
    .header {
        flex-direction: column;
        gap: 10px;
    }
    .main-content {
        flex-direction: column;
    }
}
```

Q2: 4-Column Product Gallery (CSS Grid)

index.html

HTML

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <title>Product Gallery - Q2</title>
  <link rel="stylesheet" href="style.css">
</head>
<body>
  <main class="gallery-container">
    <div class="product-card"><h3>Product 1</h3><p>$19.99</p></div>
    <div class="product-card"><h3>Product 2</h3><p>$29.99</p></div>
    <div class="product-card"><h3>Product 3</h3><p>$39.99</p></div>
    <div class="product-card"><h3>Product 4</h3><p>$49.99</p></div>
    <div class="product-card"><h3>Product 5</h3><p>$59.99</p></div>
    <div class="product-card"><h3>Product 6</h3><p>$69.99</p></div>
    <div class="product-card"><h3>Product 7</h3><p>$79.99</p></div>
    <div class="product-card"><h3>Product 8</h3><p>$89.99</p></div>
  </main>
</body>
</html>
```

style.css

CSS

```
body {
  font-family: Arial, sans-serif;
```

```
    background-color: #f9f9f9;
    padding: 20px;
}

.gallery-container {
    display: grid;
    grid-template-columns: repeat(4, 1fr);
    gap: 20px;
    max-width: 1200px;
    margin: 0 auto;
}

.product-card {
    background: white;
    padding: 30px;
    border: 1px solid #ddd;
    border-radius: 6px;
    text-align: center;
    display: flex;
    flex-direction: column;
    justify-content: space-between;
    height: 150px;
    transition: transform 0.3s ease;
}

.product-card:hover {
```

```
    transform: scale(1.05);
}

@media (max-width: 900px) {
    .gallery-container {
        grid-template-columns: repeat(2, 1fr);
    }
}

@media (max-width: 600px) {
    .gallery-container {
        grid-template-columns: 1fr;
    }
}
```

Q3: 3-Section Homepage (Flexbox)

index.html

HTML

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <title>Homepage - Q3</title>
    <link rel="stylesheet" href="style.css">
</head>
<body>
    <header class="header">
```

```
        <a href="#">Home</a>
        <a href="#">About</a>
        <a href="#">Contact</a>
    </header>

    <main class="main-area">
        <div class="box">Services</div>
        <div class="box">Projects</div>
        <div class="box">Contact</div>
    </main>

    <footer class="footer">
        <p>&copy; 2026 Company Website</p>
    </footer>
</body>
</html>
```

style.css

CSS

```
* {
    margin: 0;
    padding: 0;
    box-sizing: border-box;
}

body {
    font-family: Arial, sans-serif;
```

```
    display: flex;
    flex-direction: column;
    min-height: 100vh;
}

.header {
    display: flex;
    justify-content: space-around;
    background-color: #2c3e50;
    padding: 20px;
}

.header a {
    color: white;
    text-decoration: none;
    font-weight: bold;
}

.main-area {
    display: flex;
    justify-content: center;
    align-items: center;
    flex: 1;
    padding: 40px;
    gap: 30px;
    background-color: #ecf0f1;
```

```
    flex-wrap: wrap;
}

.box {
    background: #3498db;
    color: white;
    width: 200px;
    height: 150px;
    display: flex;
    justify-content: center;
    align-items: center;
    font-size: 20px;
    border-radius: 8px;
    transition: transform 0.3s ease;
}

.box:hover {
    transform: rotate(2deg) scale(1.02);
}

.footer {
    background-color: #2c3e50;
    color: white;
    text-align: center;
    padding: 20px;
    width: 100%;
```

```
}
```

Q4: Dashboard Layout (CSS Grid)

index.html

HTML

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <title>Dashboard - Q4</title>
  <link rel="stylesheet" href="style.css">
</head>
<body>
  <div class="dashboard-container">
    <header class="dash-header">Dashboard Header</header>
    <aside class="dash-sidebar">
      <nav>
        <a href="#">Overview</a>
        <a href="#">Analytics</a>
        <a href="#">Settings</a>
      </nav>
    </aside>
    <main class="dash-main">
      <div class="dash-box span-two">Analytics Summary</div>
      <div class="dash-box">Recent Users</div>
      <div class="dash-box">Server Load</div>
    </main>
```

```
    </div>
</body>
</html>
```

style.css

CSS

```
body {
    margin: 0;
    font-family: Arial, sans-serif;
    background-color: #f4f6f9;
}

.dashboard-container {
    display: grid;
    grid-template-columns: 220px 1fr;
    grid-template-rows: 70px 1fr;
    min-height: 100vh;
    gap: 15px;
    padding: 15px;
}

.dash-header {
    grid-column: span 2;
    background: #2c3e50;
    color: white;
    display: flex;
    align-items: center;
```

```
    padding: 0 20px;
    font-size: 20px;
    border-radius: 6px;
}

.dash-sidebar {
    background: white;
    padding: 20px;
    border-radius: 6px;
    box-shadow: 0 2px 5px rgba(0,0,0,0.05);
}

.dash-sidebar nav {
    display: flex;
    flex-direction: column;
    gap: 15px;
}

.dash-sidebar a {
    text-decoration: none;
    color: #333;
}

.dash-main {
    display: grid;
    grid-template-columns: repeat(2, 1fr);
```

```
    gap: 15px;
}

.dash-box {
    background: white;
    padding: 30px;
    border-radius: 6px;
    box-shadow: 0 2px 5px rgba(0,0,0,0.05);
    display: flex;
    justify-content: center;
    align-items: center;
    font-weight: bold;
    transition: transform 0.3s ease, box-shadow 0.3s ease;
}

.span-two {
    grid-column: span 2;
}

.dash-box:hover {
    transform: translateY(-5px);
    box-shadow: 0 10px 20px rgba(0,0,0,0.1);
}
```

Q5: Pricing Section (Flexbox & Keyframes)

index.html

HTML

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <title>Pricing - Q5</title>
  <link rel="stylesheet" href="style.css">
</head>
<body>
  <main class="pricing-container">
    <div class="price-card">
      <h3>Basic</h3>
      <p class="price">$10/mo</p>
    </div>
    <div class="price-card highlight">
      <h3>Pro</h3>
      <p class="price">$29/mo</p>
    </div>
    <div class="price-card">
      <h3>Enterprise</h3>
      <p class="price">$59/mo</p>
    </div>
  </main>
</body>
</html>
```

style.css

CSS

```
body {
    font-family: Arial, sans-serif;
    background-color: #f8f9fa;
    display: flex;
    justify-content: center;
    align-items: center;
    min-height: 100vh;
    margin: 0;
}

.pricing-container {
    display: flex;
    justify-content: center;
    align-items: center;
    gap: 20px;
    animation: fadeIn 1s ease-in-out;
}

.price-card {
    background: white;
    padding: 40px 30px;
    border-radius: 8px;
    text-align: center;
    border: 1px solid #dee2e6;
    width: 250px;
    transition: transform 0.3s ease;
```

```
}

.price-card.highlight {
    background: #007bff;
    color: white;
    transform: scale(1.05);
}

.price-card:hover {
    transform: translateY(-10px);
}

.price-card.highlight:hover {
    transform: scale(1.05) translateY(-10px);
}

@keyframes fadeIn {
    from {
        opacity: 0;
        transform: translateY(20px);
    }
    to {
        opacity: 1;
        transform: translateY(0);
    }
}
```

```
@media (max-width: 750px) {
  .pricing-container {
    flex-direction: column;
  }
  .price-card.highlight {
    transform: none;
  }
}
```

Q6: Responsive Blog Layout (CSS Grid Template Areas)

index.html

HTML

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <title>Blog Layout - Q6</title>
  <link rel="stylesheet" href="style.css">
</head>
<body>
  <div class="blog-layout">
    <header class="blog-header"><h1>My Blog</h1></header>
    <aside class="blog-sidebar">
      <h3>Categories</h3>
      <ul>
        <li>Tech</li>
```

```
            <li>Design</li>
            <li>Lifestyle</li>
        </ul>
    </aside>
    <main class="blog-main">
        <article class="post">Post 1 Content</article>
        <article class="post">Post 2 Content</article>
    </main>
    <footer class="blog-footer">Contact: info@blog.com</footer>
  </div>
</body>
</html>
```

style.css

CSS

```
body {
  margin: 0;
  font-family: Arial, sans-serif;
  background: #f0f2f5;
}

.blog-layout {
  display: grid;
  grid-template-areas:
    "header header"
    "sidebar main"
    "footer footer";
```

```
  grid-template-columns: 250px 1fr;
  grid-template-rows: auto 1fr auto;
  gap: 20px;
  padding: 20px;
  max-width: 1100px;
  margin: 0 auto;
  min-height: 100vh;
}

.blog-header { grid-area: header; background: #333; color: white; padding: 20px; border-radius: 6px; }
.blog-sidebar { grid-area: sidebar; background: white; padding: 20px; border-radius: 6px; }
.blog-main {
  grid-area: main;
  display: grid;
  grid-template-columns: repeat(2, 1fr);
  gap: 20px;
}
.post { background: white; padding: 25px; border-radius: 6px; box-shadow: 0 2px 4px rgba(0,0,0,0.05); }
.blog-footer { grid-area: footer; background: #333; color: white; padding: 15px; text-align: center; border-radius: 6px; }

@media (max-width: 800px) {
  .blog-layout {
    grid-template-areas:
      "header"
```

```
            "main"
            "sidebar"
            "footer";
        grid-template-columns: 1fr;
    }
    .blog-main {
        grid-template-columns: 1fr;
    }
}
```

Q7: CSS Loading Animation (Keyframes & Flexbox)

index.html

HTML

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <title>Loader - Q7</title>
    <link rel="stylesheet" href="style.css">
</head>
<body>
    <div class="loader-container">
        <div class="circle c1"></div>
        <div class="circle c2"></div>
        <div class="circle c3"></div>
    </div>
</body>
```

```
</html>
```

style.css

CSS

```
body {
    margin: 0;
    height: 100vh;
    display: flex;
    justify-content: center;
    align-items: center;
    background-color: #222;
}

.loader-container {
    display: flex;
    gap: 15px;
}

.circle {
    width: 20px;
    height: 20px;
    background-color: #3498db;
    border-radius: 50%;
    animation: bounce 0.6s infinite alternate;
}

.c2 {
```

```
    animation-delay: 0.2s;
}

.c3 {
    animation-delay: 0.4s;
}

@keyframes bounce {
    from {
        transform: translateY(0);
    }
    to {
        transform: translateY(-20px);
    }
}
```

Q8: Photo Gallery (Grid Auto-fit & Hover Zoom)

index.html

HTML

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <title>Photo Gallery - Q8</title>
    <link rel="stylesheet" href="style.css">
</head>
<body>
```

```
  <main class="photo-gallery">
    <figure><img src="https://via.placeholder.com/300" alt=""><figcaption>Nature 1</figcaption></figure>
    <figure><img src="https://via.placeholder.com/300" alt=""><figcaption>Nature 2</figcaption></figure>
    <figure><img src="https://via.placeholder.com/300" alt=""><figcaption>Nature 3</figcaption></figure>
    <figure><img src="https://via.placeholder.com/300" alt=""><figcaption>Nature 4</figcaption></figure>
  </main>
</body>
</html>
```

style.css

CSS

```
body {
  font-family: Arial, sans-serif;
  background: #f4f4f4;
  padding: 20px;
}

.photo-gallery {
  display: grid;
  grid-template-columns: repeat(auto-fit, minmax(200px, 1fr));
  gap: 20px;
  max-width: 1000px;
  margin: 0 auto;
}
```

```
figure {
    background: white;
    margin: 0;
    overflow: hidden;
    border-radius: 6px;
    box-shadow: 0 2px 5px rgba(0,0,0,0.1);
    text-align: center;
}

img {
    width: 100%;
    height: 180px;
    object-fit: cover;
    display: block;
    transition: transform 0.5s ease;
}

figure:hover img {
    transform: scale(1.1);
}

figcaption {
    padding: 10px;
    font-size: 14px;
    color: #333;
```

```
}
```

Q9: Card Flip Animation (3D Transforms)

index.html

HTML

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <title>Card Flip - Q9</title>
  <link rel="stylesheet" href="style.css">
</head>
<body>
  <div class="scene">
    <div class="card">
      <div class="card-face card-front">
        <h3>Front Title</h3>
        <img src="https://via.placeholder.com/150" alt="">
      </div>
      <div class="card-face card-back">
        <h3>Back Description</h3>
        <p>This is the detailed description shown on the back side of the card after flipping.</p>
      </div>
    </div>
  </div>
</body>
```

```
</html>
```

style.css

CSS

```
body {
    margin: 0;
    height: 100vh;
    display: flex;
    justify-content: center;
    align-items: center;
    background-color: #eef2f3;
}

.scene {
    width: 250px;
    height: 320px;
    perspective: 1000px;
}

.card {
    width: 100%;
    height: 100%;
    position: relative;
    transform-style: preserve-3d;
    transition: transform 0.6s ease;
}
```

```
.scene:hover .card {
    transform: rotateY(180deg);
}

.card-face {
    position: absolute;
    width: 100%;
    height: 100%;
    backface-visibility: hidden;
    display: flex;
    flex-direction: column;
    justify-content: center;
    align-items: center;
    border-radius: 8px;
    box-shadow: 0 4px 8px rgba(0,0,0,0.1);
    padding: 20px;
    box-sizing: border-box;
}

.card-front {
    background-color: white;
    color: #333;
}

.card-back {
    background-color: #3498db;
```

```
    color: white;
    transform: rotateY(180deg);
    text-align: center;
}
```

Q10: Responsive Webpage Layout (Flexbox, Grid & Keyframes)

index.html

HTML

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <title>Layout - Q10</title>
    <link rel="stylesheet" href="style.css">
</head>
<body>
    <div class="page-container">
        <header class="header">
            <div class="logo">Logo</div>
            <nav class="nav">
                <a href="#">Home</a>
                <a href="#">About</a>
            </nav>
        </header>

        <div class="content-grid">
            <main class="article">
```

```
            <h1 class="fade-title">Welcome Article</h1>
            <div class="article-boxes">
              <div class="box">Box 1</div>
              <div class="box">Box 2</div>
              <div class="box">Box 3</div>
            </div>
          </main>
          <aside class="sidebar">Sidebar Content</aside>
        </div>

        <footer class="footer">
          <p>&copy; 2026 Website</p>
        </footer>
    </div>
</body>
</html>
```

style.css

CSS

```
body {
    margin: 0;
    font-family: Arial, sans-serif;
    background: #f5f6fa;
}

.page-container {
    display: flex;
```

```
    flex-direction: column;
    min-height: 100vh;
}

.header {
    display: flex;
    justify-content: space-between;
    align-items: center;
    background: #2f3640;
    color: white;
    padding: 15px 30px;
}

.nav a {
    color: white;
    text-decoration: none;
    margin-left: 15px;
}

.content-grid {
    display: grid;
    grid-template-columns: 3fr 1fr;
    gap: 20px;
    padding: 20px;
    flex: 1;
    max-width: 1200px;
```

```
    margin: 0 auto;
    width: 100%;
    box-sizing: border-box;
}

.article {
    background: white;
    padding: 25px;
    border-radius: 6px;
}

.fade-title {
    animation: slideUp 0.8s ease-out;
}

.article-boxes {
    display: flex;
    gap: 15px;
    margin-top: 20px;
}

.article-boxes .box {
    flex: 1;
    background: #4cd137;
    color: white;
    padding: 30px;
```

```
    text-align: center;
    border-radius: 4px;
}

.sidebar {
    background: white;
    padding: 25px;
    border-radius: 6px;
}

.footer {
    background: #2f3640;
    color: white;
    text-align: center;
    padding: 15px;
}

@keyframes slideUp {
    from {
        opacity: 0;
        transform: translateY(20px);
    }
    to {
        opacity: 1;
        transform: translateY(0);
    }
```

```
}

@media (max-width: 768px) {
    .content-grid {
        grid-template-columns: 1fr;
    }
    .article-boxes {
        flex-direction: column;
    }
}
```